Day 02 DBT –C-DAC Mumbai

How to use a SELECT query to insert data from one table to another table

```
INSERT INTO student_backup
(id, name, age, course)
  SELECT
   id, name, age, course
FROM student
WHERE course = 'Java';
```

- **ALTER TABLE: ADD, MODIFY, CHANGE, DROP COLUMN, RENAME COLUMN**

```
-- 1. ADD a new column

ALTER TABLE student

ADD phone VARCHAR(15);
```

```
-- 2. MODIFY an existing column

ALTER TABLE student

MODIFY name VARCHAR(100);
```

```
-- 3. CHANGE COLUMN name and data type

ALTER TABLE student

CHANGE age student_age INT;
```

```
-- 4. DROP a column
```

ALTER TABLE student

DROP COLUMN email;

-- 5. RENAME a column

ALTER TABLE student

RENAME COLUMN phone TO mobile_no;

- DROP TABLE/DATABASE, IF EXISTS

  DROP TABLE IF EXISTS employee;

- DDL vs DML: purpose, auto-commit, rollback behaviour

To check autocommit status
We use Select @@autocommit;
For every DDL command that is create, alter or drop command autocommit is there

But in MySQL, **DDL queries cause an implicit commit**, so the ROLLBACK does not work as a way to undo the table creation.

Drop table table_name;

transactions, SAVEPOINT, and ROLLBACK are primarily used to control DML changes in MySQL. Most DDL statements cause an implicit commit, so their structural changes cannot normally be undone using ROLLBACK or SAVEPOINT.

Auto_increment

```
CREATE TABLE student (
   student_id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   age INT
);
```

Handling null and default values in table

NULL ≠ 0
NULL ≠ ''
NULL ≠ 'NULL'

NULL means the value is **unknown/not provided**.

```
CREATE TABLE student (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   age INT,
   city VARCHAR(50)
);
```

Default:

```
CREATE TABLE student (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   age INT DEFAULT 18,
   city VARCHAR(50) DEFAULT 'Pune'
);
```

How to create temporary table

```
CREATE TEMPORARY TABLE temp_employees (
   id INT PRIMARY KEY,
   name VARCHAR(50),
   department VARCHAR(50)
);
```

In case we explicitly write in the query to insert null value to that default column then it is going to insert null values but not going to replace it with default value of the column;

```
-- Insert data just like a regular table
INSERT INTO temp_employees VALUES (1, 'Alice', 'IT'), (2, 'Bob', 'HR');
```

And how to drop it ? drop table temp_employees

What is the use of a temp table?
1. Breaking down complex queries
2. Performance optimization
3. Safe data sandboxing: we can update, delete, or manipulate data from a temp table without causing any damage to main table data
4. Storing session-specific state

```
SELECT column1, column2
FROM table_name
WHERE condition;
```

Comparison Operators

The most commonly used comparison operators are:

Operator	Meaning
=	Equal to
<> or !=	Not equal to

>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Logical Operators

Logical operators allow us to combine multiple conditions.

The important ones are:

AND
OR
NOT

Tell them the meaning of the AND, OR, and NOT operators

```
SELECT *
FROM student
WHERE (course = 'Java' OR course = 'Python')
AND age > 22;
```

- UPDATE: single column, multiple columns, with WHERE, with expressions

```
UPDATE table_name
SET column_name = new_value
WHERE condition;
```

We can update multiple columns in a single UPDATE query

```
UPDATE table_name
SET column1 = value1,
   column2 = value2
WHERE condition;
```

UPDATE Using Expressions

```
UPDATE student
SET age = age + 1;
```

```
UPDATE student
SET age = age + 1
WHERE course = 'Java';
```

- Danger of UPDATE/DELETE without WHERE; safe update mode in Workbench

DELETE: selected rows, all rows

- DELETE vs TRUNCATE in detail: DML vs DDL, WHERE support, rollback, speed, AUTO_INCREMENT reset, triggers

```
TRUNCATE TABLE student;
```

```
DELETE FROM student;
      ↓
Row-level deletion
      ↓
Generally slower for very large tables

TRUNCATE TABLE student;
      ↓
Empty entire table
      ↓
Generally faster
```

DELETE can participate in an InnoDB transaction and can be rolled back before COMMIT. TRUNCATE causes an implicit commit and is not used as a rollback-able row deletion operation.

What is innodb

InnoDB is the default, general-purpose storage engine for MySQL. It is the underlying software component that MySQL uses to handle the actual creation, reading, updating, and deleting of data on your hard drive.

Feature	DELETE	TRUNCATE	DROP
Purpose	Removes rows	Removes all rows	Removes entire table
Category	DML	DDL	DDL
WHERE clause	☑Supported	❌Not supported	❌Not applicable
Delete selected rows	☑Yes	❌No	❌No
Delete all rows	☑Yes	☑Yes	☑Yes
Table structure remains?	☑Yes	☑Yes	❌No
Columns remain?	☑Yes	☑Yes	❌No
Primary key remains?	☑Yes	☑Yes	❌No, table is gone
AUTO_INCREMENT reset?	❌No	☑Yes	Table itself is deleted
Rollback	☑Possible before COMMIT	❌No normal rollback	❌No normal rollback

Speed when removing all rows	Generally slower	Generally faster	Fastest way to remove the table
DELETE triggers	☑Fires	❌Does not fire	❌Does not fire
Can use WHERE?	DELETE ... WHERE	No	No
Can reuse the table afterward?	☑Yes	☑Yes	❌Must recreate it
Data removed?	Rows	All rows	Table + rows
Best use	Remove specific records	Empty a table	Permanently remove a table

DataTypes:

In MySQL, one CHAR character does not always occupy the same amount of memory. It depends on the character set.

SHOW VARIABLES LIKE 'character_set_server';

Character Set	Maximum bytes per character
ASCII	1 byte
utf8mb3	3 bytes
utf8mb4	4 bytes

String types: CHAR vs VARCHAR, TEXT, ENUM, SET

Both CHAR and VARCHAR are used to store string/text data, but the main difference is how they handle length and storage.

Maximum length: 255 characters
n defines the fixed length of the column.
If the value is shorter than n, it is conceptually padded with trailing spaces to maintain fixed-length behavior.
When retrieved, MySQL normally removes the trailing padding spaces.
Suitable for values that are generally of a fixed or predictable length

Varchar:
Maximum declared length can be up to **65,535 bytes**, subject to MySQL's row-size and character-set limitations.
n represents the **maximum number of characters** you intend to allow.
It does **not use fixed-length padding**.
The actual value can be shorter than the declared maximum.
Suitable for values whose length **varies significantly**.

Char :
• (allows any character) (max upto 255 characters) (default width 1)
• (wastage of HD space) (searching and retrieval is very fast)
• e.g. ROLL NO, EMPNO, PANNO, etc.

Varchar :
• (allows any character) (max upto 65,535 characters) (64 KB - 1) •
(no default width) (width has to be specified) (conserve on HD space) •
(searching and retrieval is compromised) e.g. ENAME, ADDRESS, CITY, etc.

TEXT Data and Overflow Pages — Summary

In MySQL, large TEXT values may be stored in **overflow pages** by the InnoDB storage engine.

Text

Tinytext (allows any character) (max upto 255 characters)

Text (allows any character) (max upto 65,535 characters) Mediumtext (allows any character) (max upto 16,777,215 characters) (16 MB)

Longtext (allows any character) (max upto 4,294,967,295 characters) (4 GB)

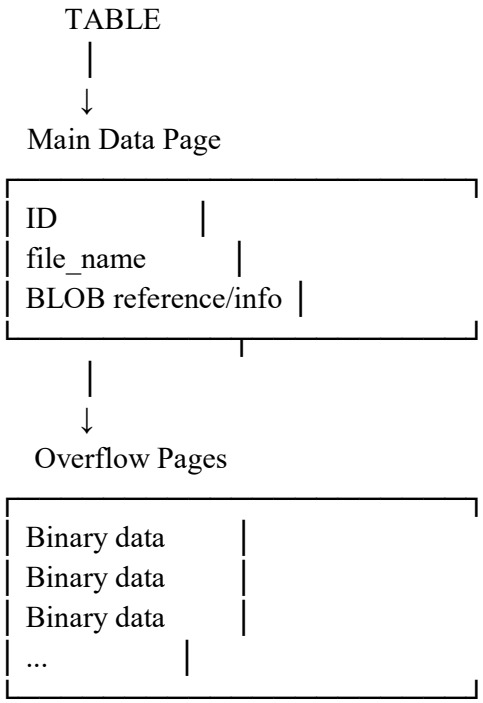
- A database table is stored in **data pages**.
- If a TEXT value is large, storing the entire value directly with the main row can make the row very large.
- InnoDB can therefore store the large portion of the TEXT value in **separate overflow pages**.
- The main row keeps information that allows InnoDB to locate the actual TEXT data.
- When a query requires the TEXT value, InnoDB accesses the required overflow page(s) and retrieves the data.
- If the query does not require the TEXT column, those overflow pages may not need to be accessed.

- The overflow pages are **still part of the same table's storage**; they are not a separate table.

Blob -> Binary Large Object
Tinyblob (max upto 255 Bytes of binary data)
Blob (max upto 65,535 Bytes of binary data)
Mediumblob (max upto 16,777,215 Bytes of binary data)
Longblob (max upto 4,294,967,295 Bytes of binary data)

BLOB can store things such as:

- Images (.jpg, .png)
- PDF files
- Audio files
- Small videos
- ZIP files
- Other binary files



Enum: Fixed set

```
CREATE TABLE student (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   gender ENUM('Male', 'Female', 'Other')
);
```

Set: it is also have multiple constant values
But we can have more than one set values

```
CREATE TABLE employee (
   id INT PRIMARY KEY AUTO_INCREMENT,
   name VARCHAR(50),
   skills SET('Java', 'Python', 'SQL', 'HTML')
);

INSERT INTO employee (name, skills)
VALUES ('Priya', 'Java,SQL,Python');
```

Integer types (Exact value) : Signed or Unsigned: - by default it is signed
Tinyint (occupies 1 Byte of storage)
Smallint (occupies 2 Bytes of storage)
Mediumint (occupies 3 Bytes of storage)
Int (occupies 4 Bytes of storage)
Bigint (occupies 8 Bytes of storage)

```
CREATE TABLE employee (
   id INT UNSIGNED
);
```

Unsigned will store only +Ve values

FLOAT

FLOAT stores a floating-point number using **4 bytes** of storage.

- Provides approximately **7 significant decimal digits of precision**.
- Requires less memory than DOUBLE.
- Suitable when very high precision is not required.
- Values are stored using an **approximate representation**, so small rounding differences can occur.

DOUBLE stores a floating-point number using **8 bytes** of storage.

- Provides approximately **15–16 significant decimal digits of precision**.
- Provides much greater precision than FLOAT.
- Requires more storage.
- Also uses an **approximate representation**.
- Suitable for scientific, engineering, and other calculations requiring higher precision.

Boolean • (True and False evaluate to 1 and 0 respectively)
e.g. MARITAL STATUS boolean *
can insert true, false, 1, or 0 *
output will display 1 or 0

Date and Time Datatypes:
Date ('YYYY-MM-DD' is the default date format)
date1-date2 -> returns number of days between the 2 dates

Time ('hh:mm:ss') or ('HHH:MM:SS')

Datetime ('YYYY-MM-DD hh:mm:ss')

Year (YYYY) (1901 to 2155)

A key difference is that MySQL TIMESTAMP values are converted between the **session time zone and UTC** when stored/retrieved.

TIMESTAMP stores **date and time**, similar to DATETIME, but it has important differences.

1970-01-01 00:00:01 UTC
to
2038-01-19 03:14:07 UTC

```
mysql> insert into student_backup select Id, Name, course from student;
Query OK, 8 rows affected (0.03 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> select * from student_backup;
+------+-----------+--------+
| Id   | Name      | course |
+------+-----------+--------+
|    1 | SAGAR     | JAVA   |
|    2 | Ritesh    | Mysql  |
|    8 | Gauri     | DotNet |
|    3 | BHAVESH   | NULL   |
|    4 | VAISHNAVI | JAVA   |
|    5 | ROHIT     | Python |
|    6 | AKSHAY    | Mysql  |
|    7 | VIDYA     | AC     |
+------+-----------+--------+
8 rows in set (0.00 sec)

mysql> delete from student_backup;
Query OK, 8 rows affected (0.05 sec)

mysql> select * from student_backup;
Empty set (0.00 sec)

mysql> insert into student_backup  select Id, Name, course from student where id >5;
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> insert into student_backup  select Id, Name, course from student where id <5;
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from student_backup;
+------+-----------+--------+
| Id   | Name      | course |
+------+-----------+--------+
|    8 | Gauri     | DotNet |
|    6 | AKSHAY    | Mysql  |
|    7 | VIDYA     | AC     |
|    1 | SAGAR     | JAVA   |
|    2 | Ritesh    | Mysql  |
|    3 | BHAVESH   | NULL   |
|    4 | VAISHNAVI | JAVA   |
+------+-----------+--------+
7 rows in set (0.00 sec)

mysql> select * from student_backup order by Id;
+------+-----------+--------+
| Id   | Name      | course |
+------+-----------+--------+
|    1 | SAGAR     | JAVA   |
|    2 | Ritesh    | Mysql  |
|    3 | BHAVESH   | NULL   |
|    4 | VAISHNAVI | JAVA   |
|    6 | AKSHAY    | Mysql  |
|    7 | VIDYA     | AC     |
|    8 | Gauri     | DotNet |
+------+-----------+--------+
7 rows in set (0.00 sec)

mysql> select * from student_backup order by Name;
```

```
mysql> select * from student_backup order by Name;
+------+-----------+--------+
| Id   | Name      | course |
+------+-----------+--------+
|    6 | AKSHAY    | Mysql  |
|    3 | BHAVESH   | NULL   |
|    8 | Gauri     | DotNet |
|    2 | Ritesh    | Mysql  |
|    1 | SAGAR     | JAVA   |
|    4 | VAISHNAVI | JAVA   |
|    7 | VIDYA     | AC     |
+------+-----------+--------+
7 rows in set (0.00 sec)

mysql> select * from student_backup order by course
+------+-----------+--------+
| Id   | Name      | course |
+------+-----------+--------+
|    3 | BHAVESH   | NULL   |
|    7 | VIDYA     | AC     |
|    8 | Gauri     | DotNet |
|    1 | SAGAR     | JAVA   |
|    4 | VAISHNAVI | JAVA   |
|    6 | AKSHAY    | Mysql  |
|    2 | Ritesh    | Mysql  |
+------+-----------+--------+
7 rows in set (0.00 sec)
```

```
mysql> alter table student
    -> add column grade char(5);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| id          | int         | YES  |     | NULL    |       |
| Name        | varchar(30) | YES  |     | NULL    |       |
| student_age | char(10)    | YES  |     | NULL    |       |
| course      | char(10)    | YES  |     | NULL    |       |
| City        | varchar(10) | YES  |     | NULL    |       |
| grade       | char(5)     | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)

mysql> alter table student
    -> add column emailid char(5);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| id          | int         | YES  |     | NULL    |       |
| Name        | varchar(30) | YES  |     | NULL    |       |
| student_age | char(10)    | YES  |     | NULL    |       |
| course      | char(10)    | YES  |     | NULL    |       |
| City        | varchar(10) | YES  |     | NULL    |       |
| grade       | char(5)     | YES  |     | NULL    |       |
| emailid     | char(5)     | YES  |     | NULL    |       |
```

```
mysql> alter table student
    -> modify emailid varchar(30);
Query OK, 8 rows affected (0.16 sec)
Records: 8  Duplicates: 0  Warnings: 0

mysql> desc student;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| id          | int         | YES  |     | NULL    |       |
| Name        | varchar(30) | YES  |     | NULL    |       |
| student_age | char(10)    | YES  |     | NULL    |       |
| course      | char(10)    | YES  |     | NULL    |       |
| City        | varchar(10) | YES  |     | NULL    |       |
| grade       | char(5)     | YES  |     | NULL    |       |
| emailid     | varchar(30) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

```
mysql>
mysql> alter table student
    -> change Name Student_name varchar(40);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| id           | int         | YES  |     | NULL    |       |
| Student_name | varchar(40) | YES  |     | NULL    |       |
| student_age  | char(10)    | YES  |     | NULL    |       |
| course       | char(10)    | YES  |     | NULL    |       |
| City         | varchar(10) | YES  |     | NULL    |       |
| grade        | char(5)     | YES  |     | NULL    |       |
| emailid      | varchar(30) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
7 rows in set (0.01 sec)
```

```
mysql>
mysql> alter table student
    -> drop column grade;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| id           | int         | YES  |     | NULL    |       |
| Student_name | varchar(40) | YES  |     | NULL    |       |
| student_age  | char(10)    | YES  |     | NULL    |       |
| course       | char(10)    | YES  |     | NULL    |       |
| City         | varchar(10) | YES  |     | NULL    |       |
| emailid      | varchar(30) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

```
mysql> drop table demo;
Query OK, 0 rows affected (0.07 sec

mysql> show tables;
+--------------------+
| Tables_in_aug25    |
+--------------------+
| student            |
| student4           |
| student_1          |
| student_backup     |
| student_delete_log |
| xyz_student        |
+--------------------+
6 rows in set (0.01 sec)
```

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student;
+------+--------------+-------------+--------+----------+---------+
| id   | Student_name | student_age | course | City     | emailid |
+------+--------------+-------------+--------+----------+---------+
|    1 | SAGAR        | 30          | JAVA   | Mumbai   | NULL    |
|    2 | Ritesh       | 25          | Mysql  | Satara   | NULL    |
|    8 | Gauri        | 25          | DotNet | Pune     | NULL    |
|    5 | ROHIT        | 28          | Python | Kolhapur | NULL    |
|    6 | AKSHAY       | 30          | Mysql  | Sangli   | NULL    |
|    7 | VIDYA        | 25          | AC     | PUNE     | NULL    |
+------+--------------+-------------+--------+----------+---------+
6 rows in set (0.00 sec)

mysql> savepoint s1;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into student (id, student_name, student_age, course, city, emailid) values (9,'Sammed', '30',
    -> insert into student (id, student_name, student_age, course, city, emailid) values (9,'Sammed', '30', ^C
mysql> insert into student (id, student_name, student_age, course, city, emailid) values (9,'Sammed', '30', 'msc', 'Kolhapur' ,'sammed@gmail.com');
Query OK, 1 row affected (0.00 sec)

mysql> select * from student;
+------+--------------+-------------+--------+----------+------------------+
| id   | Student_name | student_age | course | City     | emailid          |
+------+--------------+-------------+--------+----------+------------------+
|    1 | SAGAR        | 30          | JAVA   | Mumbai   | NULL             |
|    2 | Ritesh       | 25          | Mysql  | Satara   | NULL             |
|    8 | Gauri        | 25          | DotNet | Pune     | NULL             |
|    5 | ROHIT        | 28          | Python | Kolhapur | NULL             |
|    6 | AKSHAY       | 30          | Mysql  | Sangli   | NULL             |
|    7 | VIDYA        | 25          | AC     | PUNE     | NULL             |
```

```
mysql> savepoint s2;
Query OK, 0 rows affected (0.00 sec)

mysql> insert into student (id, student_name, student_age, course, city, emailid) values (10,'Yash', '28', 'BSC', 'Nagpur' ,'Yash@gmail.com');
Query OK, 1 row affected (0.00 sec)

mysql> select * from student;
+------+--------------+-------------+--------+----------+------------------+
| id   | Student_name | student_age | course | City     | emailid          |
+------+--------------+-------------+--------+----------+------------------+
|    1 | SAGAR        | 30          | JAVA   | Mumbai   | NULL             |
|    2 | Ritesh       | 25          | Mysql  | Satara   | NULL             |
|    8 | Gauri        | 25          | DotNet | Pune     | NULL             |
|    5 | ROHIT        | 28          | Python | Kolhapur | NULL             |
|    6 | AKSHAY       | 30          | Mysql  | Sangli   | NULL             |
|    7 | VIDYA        | 25          | AC     | PUNE     | NULL             |
|    9 | Sammed       | 30          | msc    | Kolhapur | sammed@gmail.com |
|   10 | Yash         | 28          | BSC    | Nagpur   | Yash@gmail.com   |
+------+--------------+-------------+--------+----------+------------------+
8 rows in set (0.00 sec)

mysql> rollback to s2;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student;
+------+--------------+-------------+--------+----------+------------------+
| id   | Student_name | student_age | course | City     | emailid          |
+------+--------------+-------------+--------+----------+------------------+
|    1 | SAGAR        | 30          | JAVA   | Mumbai   | NULL             |
|    2 | Ritesh       | 25          | Mysql  | Satara   | NULL             |
|    8 | Gauri        | 25          | DotNet | Pune     | NULL             |
|    5 | ROHIT        | 28          | Python | Kolhapur | NULL             |
|    6 | AKSHAY       | 30          | Mysql  | Sangli   | NULL             |
|    7 | VIDYA        | 25          | AC     | PUNE     | NULL             |
```

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> delete from student where id='2';
Query OK, 1 row affected (0.00 sec)

mysql> select * from student order by id;
+------+--------------+-------------+---------+-----------+---------+
| id   | Student_name | student_age | course  | City      | emailid |
+------+--------------+-------------+---------+-----------+---------+
|    1 | SAGAR        | 30          | JAVA    | Kharghar1 | NULL    |
|    5 | ROHIT        | 28          | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 30          | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 25          | PGCP-AC | PUNE      | NULL    |
|    8 | Gauri        | 25          | DotNet  | Pune      | NULL    |
+------+--------------+-------------+---------+-----------+---------+
5 rows in set (0.00 sec)

mysql> rollback;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student order by id;
+------+--------------+-------------+---------+-----------+---------+
| id   | Student_name | student_age | course  | City      | emailid |
+------+--------------+-------------+---------+-----------+---------+
|    1 | SAGAR        | 30          | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 25          | Mysql   | Satara    | NULL    |
|    5 | ROHIT        | 28          | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 30          | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 25          | PGCP-AC | PUNE      | NULL    |
|    8 | Gauri        | 25          | DotNet  | Pune      | NULL    |
+------+--------------+-------------+---------+-----------+---------+
6 rows in set (0.00 sec)
```

```
mysql> start transaction;
Query OK, 0 rows affected (0.00 sec)

mysql> delete from student;
Query OK, 6 rows affected (0.00 sec)

mysql> select * from student order by id;
Empty set (0.00 sec)

mysql> rollback;
Query OK, 0 rows affected (0.00 sec)

mysql> select * from student order by id;
+------+--------------+--------------+----------+-----------+----------+
| id   | Student_name | student_age | course   | City      | emailid  |
+------+--------------+--------------+----------+-----------+----------+
|    1 | SAGAR        | 30           | JAVA     | Kharghar1 | NULL     |
|    2 | Ritesh       | 25           | Mysql    | Satara    | NULL     |
|    5 | ROHIT        | 28           | Adjava   | Kolhapur  | NULL     |
|    6 | AKSHAY       | 30           | Mysql    | Sangli    | NULL     |
|    7 | VIDYA        | 25           | PGCP-AC  | PUNE      | NULL     |
|    8 | Gauri        | 25           | DotNet   | Pune      | NULL     |
+------+--------------+--------------+----------+-----------+----------+
6 rows in set (0.00 sec)

mysql> select @@autocommit;
+---------------+
| @@autocommit |
+---------------+
|             1 |
+---------------+
1 row in set (0.00 sec)
```

```
mysql> create table Employee ( EmpId Int  Primary key auto_increment, EmpName varchar(25), deptno Int, salary Int, emailid varchar(25));
Query OK, 0 rows affected (0.03 sec)

mysql> desc employee;
+---------+-------------+------+-----+---------+----------------+
| Field   | Type        | Null | Key | Default | Extra          |
+---------+-------------+------+-----+---------+----------------+
| EmpId   | int         | NO   | PRI | NULL    | auto_increment |
| EmpName | varchar(25) | YES  |     | NULL    |                |
| deptno  | int         | YES  |     | NULL    |                |
| salary  | int         | YES  |     | NULL    |                |
| emailid | varchar(25) | YES  |     | NULL    |                |
+---------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> insert into employee ( empname,deptno,salary, emailid )
    -> values
    -> ('ANIKET', 20, 50000, 'aniket@gmail.com'),
    -> ('Sahil', 25, 55000, 'sahil@gmail.com'),
    -> ('Aditya', 30, 60000, 'aditya@gmail.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from employee;
+-------+---------+--------+--------+------------------+
| EmpId | EmpName | deptno | salary | emailid          |
+-------+---------+--------+--------+------------------+
|     1 | ANIKET  |     20 |  50000 | aniket@gmail.com |
|     2 | Sahil   |     25 |  55000 | sahil@gmail.com  |
|     3 | Aditya  |     30 |  60000 | aditya@gmail.com |
+-------+---------+--------+--------+------------------+
3 rows in set (0.00 sec)
```

```
mysql> insert into employee (empid, empname, deptno, salary, emailid)
    -> values (4, 'Ritik', 30, 65000, 'ritik@gmail.com');
Query OK, 1 row affected (0.01 sec)

mysql> select * from employee;
+-------+---------+--------+--------+-------------------+
| EmpId | EmpName | deptno | salary | emailid           |
+-------+---------+--------+--------+-------------------+
|     1 | ANIKET  |     20 |  50000 | aniket@gmail.com  |
|     2 | Sahil   |     25 |  55000 | sahil@gmail.com   |
|     3 | Aditya  |     30 |  60000 | aditya@gmail.com  |
|     4 | Ritik   |     30 |  65000 | ritik@gmail.com   |
+-------+---------+--------+--------+-------------------+
4 rows in set (0.00 sec)

mysql> insert into employee (empid, empname, deptno, salary, emailid)
    -> values (10, 'Vishwa', 25, 60000, 'Vishwa@gmail.com');
Query OK, 1 row affected (0.00 sec)

mysql> select * from employee;
+-------+---------+--------+--------+-------------------+
| EmpId | EmpName | deptno | salary | emailid           |
+-------+---------+--------+--------+-------------------+
|     1 | ANIKET  |     20 |  50000 | aniket@gmail.com  |
|     2 | Sahil   |     25 |  55000 | sahil@gmail.com   |
|     3 | Aditya  |     30 |  60000 | aditya@gmail.com  |
|     4 | Ritik   |     30 |  65000 | ritik@gmail.com   |
|    10 | Vishwa  |     25 |  60000 | Vishwa@gmail.com  |
+-------+---------+--------+--------+-------------------+
5 rows in set (0.00 sec)
```

```
mysql> insert into employee (empid, empname, deptno, salary, emailid)
    -> values (10, 'Milind', 25, 60000, 'Milind@gmail.com');
ERROR 1062 (23000): Duplicate entry '10' for key 'employee.PRIMARY'
mysql> desc employee;
+---------+-------------+------+-----+---------+----------------+
| Field   | Type        | Null | Key | Default | Extra          |
+---------+-------------+------+-----+---------+----------------+
| EmpId   | int         | NO   | PRI | NULL    | auto_increment |
| EmpName | varchar(25) | YES  |     | NULL    |                |
| deptno  | int         | YES  |     | NULL    |                |
| salary  | int         | YES  |     | NULL    |                |
| emailid | varchar(25) | YES  |     | NULL    |                |
+---------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql> create table department ( deptid primary key, deptname varchar(25), salary int default 25000, location varchar default 'Kharghar');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use
 near 'primary key, deptname varchar(25), salary int default 25000, location varchar de' at line 1
mysql> create table department ( deptid primary key, deptname varchar(25), salary int default 25000, location varchar(40) default 'Kharghar');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use
 near 'primary key, deptname varchar(25), salary int default 25000, location varchar(40' at line 1
mysql> create table department ( deptid int primary key, deptname varchar(25), salary int default 25000, location varchar(40) default 'Kharghar');
Query OK, 0 rows affected (0.02 sec)

mysql> desc department;
+----------+-------------+------+-----+----------+-------+
| Field    | Type        | Null | Key | Default  | Extra |
+----------+-------------+------+-----+----------+-------+
| deptid   | int         | NO   | PRI | NULL     |       |
| deptname | varchar(25) | YES  |     | NULL     |       |
| salary   | int         | YES  |     | 25000    |       |
| location | varchar(40) | YES  |     | Kharghar |       |
+----------+-------------+------+-----+----------+-------+
4 rows in set (0.00 sec)
```

```
mysql> insert into department (deptId, deptname, salary, location)
    -> values
    -> (1, 'Education', 50000, 'Juhu'),
    -> (2, 'Research', 60000, 'Bandra');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from employee;
+-------+---------+--------+--------+------------------+
| EmpId | EmpName | deptno | salary | emailid          |
+-------+---------+--------+--------+------------------+
|     1 | ANIKET  |     20 |  50000 | aniket@gmail.com |
|     2 | Sahil   |     25 |  55000 | sahil@gmail.com  |
|     3 | Aditya  |     30 |  60000 | aditya@gmail.com |
|     4 | Ritik   |     30 |  65000 | ritik@gmail.com  |
|    10 | Vishwa  |     25 |  60000 | Vishwa@gmail.com |
|    11 | Shilsir |     20 |  50000 | Shil@gmail.com   |
+-------+---------+--------+--------+------------------+
6 rows in set (0.00 sec)

mysql> select * from department;
+--------+------------+--------+----------+
| deptid | deptname   | salary | location |
+--------+------------+--------+----------+
|      1 | Education  |  50000 | Juhu     |
|      2 | Research   |  60000 | Bandra   |
+--------+------------+--------+----------+
2 rows in set (0.00 sec)
```

```
mysql> create temporary table employee_temp ( empId Int, empName varchar(20), salary Int, DeptName varchar(20));
Query OK, 0 rows affected (0.00 sec)

mysql> desc employee_temp;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| empId    | int         | YES  |     | NULL    | NULL  |
| empName  | varchar(20) | YES  |     | NULL    | NULL  |
| salary   | int         | YES  |     | NULL    | NULL  |
| DeptName | varchar(20) | YES  |     | NULL    | NULL  |
+----------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> insert into employee_temp (1, 'Shweta' 60000, 'Education'),
    -> (2,'Vipul', 70000, 'Training');
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use
 near '1, 'Shweta' 60000, 'Education'),
(2,'Vipul', 70000, 'Training')' at line 1
mysql> (2,'Vipul', 70000, 'Training');^C
mysql>
mysql> insert into employee_temp ( empId, empName, salary, DeptName)
    -> values(1, 'Shweta', 50000, 'Education'),
    -> (2, 'Vipul' , 60000, 'Training');
Query OK, 2 rows affected (0.00 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from employee_temp;
+-------+---------+--------+-----------+
| empId | empName | salary | DeptName  |
+-------+---------+--------+-----------+
|     1 | Shweta  |  50000 | Education |
|     2 | Vipul   |  60000 | Training  |
+-------+---------+--------+-----------+
```

```
mysql> drop table employee_temp;
Query OK, 0 rows affected (0.00 sec)

mysql> show tabls;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use
 near 'tabls' at line 1
mysql> show tables;
+--------------------+
| Tables_in_aug25    |
+--------------------+
| demo               |
| department         |
| employee           |
| student            |
| student4           |
| student_1          |
| student_backup     |
| student_delete_log |
| xyz_student        |
+--------------------+
9 rows in set (0.00 sec)

mysql> select * from student;
+------+--------------+-------------+---------+-----------+---------+
| id   | Student_name | student_age | course  | City      | emailid |
+------+--------------+-------------+---------+-----------+---------+
|    1 | SAGAR        | 30          | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 25          | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 25          | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 28          | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 30          | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 25          | PGCP-AC | PUNE      | NULL    |
+------+--------------+-------------+---------+-----------+---------+
6 rows in set (0.00 sec)
```

```
mysql> update student set student_age = student_age+2 where id=6;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from student;
+------+--------------+-------------+---------+-----------+---------+
| id   | Student_name | student_age | course  | City      | emailid |
+------+--------------+-------------+---------+-----------+---------+
|    1 | SAGAR        | 30          | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 25          | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 25          | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 28          | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 32          | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 25          | PGCP-AC | PUNE      | NULL    |
+------+--------------+-------------+---------+-----------+---------+
6 rows in set (0.00 sec)

mysql> update student set student_age = student_age+2;
Query OK, 6 rows affected (0.00 sec)
Rows matched: 6  Changed: 6  Warnings: 0

mysql> select * from student;
+------+--------------+-------------+---------+-----------+---------+
| id   | Student_name | student_age | course  | City      | emailid |
+------+--------------+-------------+---------+-----------+---------+
|    1 | SAGAR        | 32          | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 27          | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 27          | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 30          | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 34          | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 27          | PGCP-AC | PUNE      | NULL    |
+------+--------------+-------------+---------+-----------+---------+
6 rows in set (0.00 sec)
```

```
mysql> drop table if exists xyz_student;
Query OK, 0 rows affected (0.01 sec)

mysql> drop table if exists xyz_student;
Query OK, 0 rows affected, 1 warning (0.01 sec)

mysql>
mysql> desc student;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| id           | int         | YES  |     | NULL    |       |
| Student_name | varchar(40) | YES  |     | NULL    |       |
| student_age  | char(10)    | YES  |     | NULL    |       |
| course       | char(10)    | YES  |     | NULL    |       |
| City         | varchar(10) | YES  |     | NULL    |       |
| emailid      | varchar(30) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
6 rows in set (0.00 sec)

mysql> alter table student rename column studet_age to age;
ERROR 1054 (42S22): Unknown column 'studet_age' in 'student'
mysql> alter table student rename column student_age to age;
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| id           | int         | YES  |     | NULL    |       |
| Student_name | varchar(40) | YES  |     | NULL    |       |
| age          | char(10)    | YES  |     | NULL    |       |
| course       | char(10)    | YES  |     | NULL    |       |
```

```
mysql> rename table student to pgcpstudent;
Query OK, 0 rows affected (0.01 sec)

mysql> show tables;
+--------------------+
| Tables_in_aug25    |
+--------------------+
| department         |
| employee           |
| pgcpstudent        |
| student4           |
| student_1          |
| student_backup     |
| student_delete_log |
+--------------------+
7 rows in set (0.00 sec)
```

```
mysql> select * from pgcpstudent where course='JAVA';
+------+--------------+------+--------+----------+---------+
| id   | Student_name | age  | course | City     | emailid |
+------+--------------+------+--------+----------+---------+
|    1 | SAGAR        | 32   | JAVA   | Kharghar1 | NULL    |
+------+--------------+------+--------+----------+---------+
1 row in set (0.00 sec)

mysql> select * from pgcpstudent where course!='JAVA';
+------+--------------+------+---------+----------+---------+
| id   | Student_name | age  | course  | City     | emailid |
+------+--------------+------+---------+----------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara   | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune     | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur | NULL    |
|    6 | AKSHAY       | 34   | Mysql   | Sangli   | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE     | NULL    |
+------+--------------+------+---------+----------+---------+
5 rows in set (0.00 sec)

mysql> select * from pgcpstudent where course <>'JAVA';
+------+--------------+------+---------+----------+---------+
| id   | Student_name | age  | course  | City     | emailid |
+------+--------------+------+---------+----------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara   | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune     | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur | NULL    |
|    6 | AKSHAY       | 34   | Mysql   | Sangli   | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE     | NULL    |
+------+--------------+------+---------+----------+---------+
5 rows in set (0.00 sec)

mysql> select * from pgcpstudent where age > 30;
```

```
mysql> select * from pgcpstudent where age < 30;
+------+--------------+------+---------+--------+---------+
| id   | Student_name | age  | course  | City   | emailid |
+------+--------------+------+---------+--------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune   | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE   | NULL    |
+------+--------------+------+---------+--------+---------+
3 rows in set (0.00 sec)

mysql> select * from pgcpstudent where age >= 30;
+------+--------------+------+--------+-----------+---------+
| id   | Student_name | age  | course | City      | emailid |
+------+--------------+------+--------+-----------+---------+
|    1 | SAGAR        | 32   | JAVA   | Kharghar1 | NULL    |
|    5 | ROHIT        | 30   | Adjava | Kolhapur  | NULL    |
|    6 | AKSHAY       | 34   | Mysql  | Sangli    | NULL    |
+------+--------------+------+--------+-----------+---------+
3 rows in set (0.00 sec)

mysql> select * from pgcpstudent where age <= 30;
+------+--------------+------+---------+----------+---------+
| id   | Student_name | age  | course  | City     | emailid |
+------+--------------+------+---------+----------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara   | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune     | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE     | NULL    |
+------+--------------+------+---------+----------+---------+
4 rows in set (0.00 sec)

mysql> select * from pgcpstudent where course='JAVA';
+------+--------------+------+--------+-----------+---------+
```

```
mysql> select * from pgcpstudent where age <=30 AND course='Mysql';
+------+--------------+------+--------+--------+---------+
| id   | Student_name | age  | course | City   | emailid |
+------+--------------+------+--------+--------+---------+
|    2 | Ritesh       | 27   | Mysql  | Satara | NULL    |
+------+--------------+------+--------+--------+---------+
1 row in set (0.00 sec)

mysql> select * from pgcpstudent where age <=30 and course='Mysql';
+------+--------------+------+--------+--------+---------+
| id   | Student_name | age  | course | City   | emailid |
+------+--------------+------+--------+--------+---------+
|    2 | Ritesh       | 27   | Mysql  | Satara | NULL    |
+------+--------------+------+--------+--------+---------+
1 row in set (0.00 sec)

mysql> select * from pgcpstudent where age <=30 or course='Mysql';
+------+--------------+------+---------+----------+---------+
| id   | Student_name | age  | course  | City     | emailid |
+------+--------------+------+---------+----------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara   | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune     | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur | NULL    |
|    6 | AKSHAY       | 34   | Mysql   | Sangli   | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE     | NULL    |
+------+--------------+------+---------+----------+---------+
5 rows in set (0.00 sec)

mysql> select * from pgcpstudent where emailid = null;
Empty set (0.00 sec)

mysql> select * from pgcpstudent where emailid ='';
Empty set (0.00 sec)
```

```
mysql> select * from pgcpstudent where emailid is null;
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    1 | SAGAR        | 32   | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 27   | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 34   | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
6 rows in set (0.00 sec)

mysql> select * from pgcpstudent where emailid is null;
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    1 | SAGAR        | 32   | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 27   | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur  | NULL    |
|    6 | AKSHAY       | 34   | Mysql   | Sangli    | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
6 rows in set (0.00 sec)

mysql> update pgcpstudent set emailid='akshay@gmail.com' where id=6;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> select * from pgcpstudent where emailid is null;
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    1 | SAGAR        | 32   | JAVA    | Kharghar1 | NULL    |
|    2 | Ritesh       | 27   | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur  | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
5 rows in set (0.00 sec)

mysql> select * from pgcpstudent where emailid is not null;
+------+--------------+------+--------+--------+------------------+
| id   | Student_name | age  | course | City   | emailid          |
+------+--------------+------+--------+--------+------------------+
|    6 | AKSHAY       | 34   | Mysql  | Sangli | akshay@gmail.com |
+------+--------------+------+--------+--------+------------------+
1 row in set (0.00 sec)

mysql> select * from pgcpstudent where age =27 And emailid is null;
+------+--------------+------+---------+--------+---------+
| id   | Student_name | age  | course  | City   | emailid |
+------+--------------+------+---------+--------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune   | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE   | NULL    |
+------+--------------+------+---------+--------+---------+
3 rows in set (0.00 sec)

mysql> select * from pgcpstudent where age =27 And emailid is null and city = 'PUNE';
+------+--------------+------+---------+------+---------+
| id   | Student_name | age  | course  | City | emailid |
+------+--------------+------+---------+------+---------+
```

```
mysql> select * from pgcpstudent where  age is null;
+------+--------------+------+--------+------+---------+
| id   | Student_name | age  | course | City | emailid |
+------+--------------+------+--------+------+---------+
|   10 | Aniket       | NULL | DotNet | NULL | NULL    |
+------+--------------+------+--------+------+---------+
1 row in set (0.00 sec)

mysql> select * from pgcpstudent where age < 32 and city='pune' or course= 'JAVA';
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    1 | SAGAR        | 32   | JAVA    | Kharghar1 | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune      | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
3 rows in set (0.00 sec)

mysql> select * from pgcpstudent where age < 32 and city='pune'
    -> select * from pgcpstudent where age < 32 and city='pune';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use
 near 'select * from pgcpstudent where age < 32 and city='pune'' at line 2
mysql> select * from pgcpstudent where age < 32 and city ='pune';
+------+--------------+------+---------+------+---------+
| id   | Student_name | age  | course  | City | emailid |
+------+--------------+------+---------+------+---------+
|    8 | Gauri        | 27   | DotNet  | Pune | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE | NULL    |
+------+--------------+------+---------+------+---------+
2 rows in set (0.00 sec)

mysql> select * from pgcpstudent where age < 32 and city='pune' and course= 'JAVA';
Empty set (0.00 sec)
```

```
mysql> select * from pgcpstudent where age < 32 or city='pune' and course= 'JAVA';
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur  | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
4 rows in set (0.00 sec)

mysql> select * from pgcpstudent where (age < 32 ) or (city='pune' and course= 'JAVA');
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    2 | Ritesh       | 27   | Mysql   | Satara    | NULL    |
|    8 | Gauri        | 27   | DotNet  | Pune      | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur  | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
4 rows in set (0.00 sec)

mysql> select * from pgcpstudent where (age < 32  or city='pune') and (course= 'JAVA');
Empty set (0.00 sec)

mysql> select * from pgcpstudent where course in ('Mysql', 'DotNet');
+------+--------------+------+--------+--------+-------------------+
| id   | Student_name | age  | course | City   | emailid           |
+------+--------------+------+--------+--------+-------------------+
|    2 | Ritesh       | 27   | Mysql  | Satara | NULL              |
|    8 | Gauri        | 27   | DotNet | Pune   | NULL              |
|    6 | AKSHAY       | 34   | Mysql  | Sangli | akshay@gmail.com  |
|   10 | Aniket       | NULL | DotNet | NULL   | NULL              |
+------+--------------+------+--------+--------+-------------------+
```

```
mysql> select * from pgcpstudent where course Not  in ('Mysql', 'DotNet');
+------+--------------+------+---------+-----------+---------+
| id   | Student_name | age  | course  | City      | emailid |
+------+--------------+------+---------+-----------+---------+
|    1 | SAGAR        | 32   | JAVA    | Kharghar1 | NULL    |
|    5 | ROHIT        | 30   | Adjava  | Kolhapur  | NULL    |
|    7 | VIDYA        | 27   | PGCP-AC | PUNE      | NULL    |
+------+--------------+------+---------+-----------+---------+
3 rows in set (0.00 sec)

mysql> select * from pgcpstudent where course= 'Mysql' or  course ='DotNet';
+------+--------------+------+--------+--------+------------------+
| id   | Student_name | age  | course | City   | emailid          |
+------+--------------+------+--------+--------+------------------+
|    2 | Ritesh       | 27   | Mysql  | Satara | NULL             |
|    8 | Gauri        | 27   | DotNet | Pune   | NULL             |
|    6 | AKSHAY       | 34   | Mysql  | Sangli | akshay@gmail.com |
|   10 | Aniket       | NULL | DotNet | NULL   | NULL             |
+------+--------------+------+--------+--------+------------------+
4 rows in set (0.00 sec)

mysql> select * from pgcpstudent where course in ('Mysql', 'DotNet');
+------+--------------+------+--------+--------+------------------+
| id   | Student_name | age  | course | City   | emailid          |
+------+--------------+------+--------+--------+------------------+
|    2 | Ritesh       | 27   | Mysql  | Satara | NULL             |
|    8 | Gauri        | 27   | DotNet | Pune   | NULL             |
|    6 | AKSHAY       | 34   | Mysql  | Sangli | akshay@gmail.com |
|   10 | Aniket       | NULL | DotNet | NULL   | NULL             |
+------+--------------+------+--------+--------+------------------+
4 rows in set (0.00 sec)

mysql> select * from pgcpstudent where id in (1,2,3,4,5,6,7);
```

```
mysql> show variables like 'character_set_server';
+----------------------+---------+
| Variable_name        | Value   |
+----------------------+---------+
| character_set_server | utf8mb4 |
+----------------------+---------+
1 row in set (0.01 sec)

mysql> create table enumdemo( StudId Int, Name varchar(20), Gender enum('Male','Female','Other'));
Query OK, 0 rows affected (0.03 sec)

mysql> desc enumdemo;
+--------+---------------------------------+------+-----+---------+-------+
| Field  | Type                            | Null | Key | Default | Extra |
+--------+---------------------------------+------+-----+---------+-------+
| StudId | int                             | YES  |     | NULL    |       |
| Name   | varchar(20)                     | YES  |     | NULL    |       |
| Gender | enum('Male','Female','Other') | YES  |     | NULL    |       |
+--------+---------------------------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql> insert into enumdemo (studid, name, gender) values (1, 'Aniket', 'Male');
Query OK, 1 row affected (0.01 sec)

mysql> select * from enumdemo;
+--------+--------+--------+
| StudId | Name   | Gender |
+--------+--------+--------+
|      1 | Aniket | Male   |
+--------+--------+--------+
1 row in set (0.00 sec)
```

```
mysql> select * from enumdemo;
+--------+--------+--------+
| StudId | Name   | Gender |
+--------+--------+--------+
|      1 | Aniket | Male   |
|      2 | Shweta | Female |
+--------+--------+--------+
2 rows in set (0.00 sec)

mysql> create table setdemo (studId int, name varchar(20), Skills set('Java', 'Mysql', 'Python', 'Cpp'));
Query OK, 0 rows affected (0.02 sec)

mysql> desc setdemo;
+--------+-----------------------------------+------+-----+---------+-------+
| Field  | Type                              | Null | Key | Default | Extra |
+--------+-----------------------------------+------+-----+---------+-------+
| studId | int                               | YES  |     | NULL    |       |
| name   | varchar(20)                       | YES  |     | NULL    |       |
| Skills | set('Java','Mysql','Python','Cpp') | YES  |     | NULL    |       |
+--------+-----------------------------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql> insert into setdemo (studId, name, skills) values (1, 'Aniket', 'Java,Mysql,Cpp');
Query OK, 1 row affected (0.01 sec)

mysql> select * from setdemo;
+--------+--------+----------------+
| studId | name   | Skills         |
+--------+--------+----------------+
|      1 | Aniket | Java,Mysql,Cpp |
+--------+--------+----------------+
1 row in set (0.00 sec)

mysql> insert into setdemo (studId, name, skills) values (2, 'Shweta', 'JAVA,MYSQL,Cpp');
```